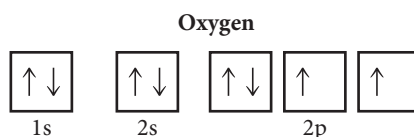


STUDENT BOOK ANSWERS

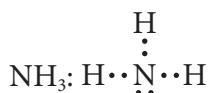
Chapter 6 Water and the intermolecular forces

Question set 6.1

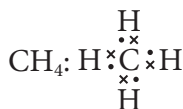
- 1
 - a 1
 - b 2
 - c 5
 - d 4
 - e 1
 - f 6
- 2 Electron configuration of oxygen is: $1s^2 2s^2 2p^4$. The outer shell has 2 electrons in the s subshell and four electrons in the p subshell. As there are three orbitals in the p subshell and electrons repel each other; there is one orbital with 2 electrons and 2 orbitals with unpaired electrons. The unpaired electrons are the ones that will form covalent bonds. There are two sets of paired electrons – the $2s^2$ and the one filled $2p^2$ orbital. These are the two lone pairs of electrons.



- 3 HCl: $\text{H} - \ddot{\text{Cl}}:$
 - a Hydrochloric acid has one bonding pair of electrons and there are three non-bonding pairs on the chlorine.
 - b Linear
 - c Diatomic molecule



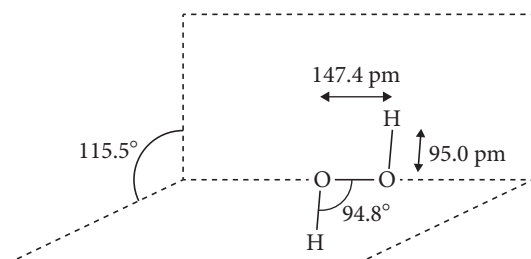
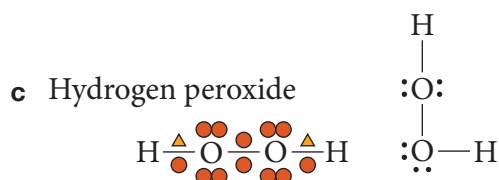
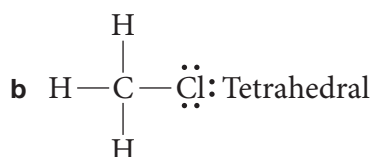
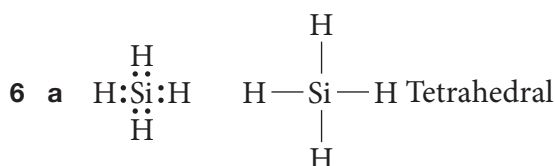
- a Ammonia has three bonding pairs and one lone pair on the nitrogen.
- b Pyramid
- c The hydrogen atoms are at three corners of the tetrahedron and the lone pair occupies the final spot. This results in a pyramid shape for ammonia.



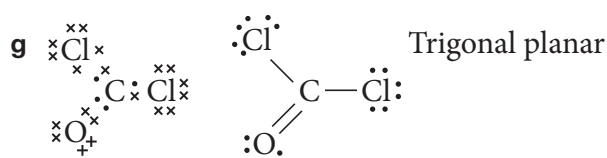
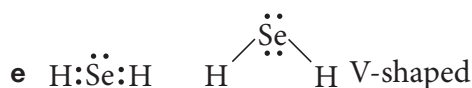
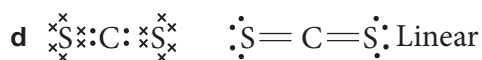
- a CH_4 has four bonding pairs of electrons.
- b Tetrahedral
- c The four hydrogen atoms and their bonding pairs of electrons are farthest apart in a tetrahedral shape.

- 4 Sulfur is the central atom in H_2S and has six electrons in the outer shell, like oxygen in water. Both will form two covalent bonds with the two hydrogen atoms. There will be two covalent bonds and two non-bonding pairs or lone pairs on the sulfur, like in oxygen. The two non-bonding pairs take up two spaces in the tetrahedral structure which results in a V-shaped molecule.
- 5 Oxygen is the central atom in water. Oxygen has two lone pairs and two bonding pairs of electrons. These four pairs are directed to the corners of the tetrahedron. Two of these are taken up by oxygen's lone pairs. This results in a V shape for water molecules.

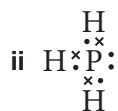
Carbon is the central atom in carbon dioxide. It forms a double bond with each oxygen atom. This is counted as two electron pairs around the central atom. There are no lone pairs on the carbon. The resultant shape is linear.



The shape around each oxygen is a V-shape but the hydrogen atoms are effectively perpendicular to each other. This maximises the distance between the electron pairs.



7 a i $1s^2 2s^2 2p^6 3s^2 3p^3$



- iii There are five valence electrons so P can form covalent bonds with 3 hydrogen atoms. P is the central atom and there is one lone pair and three bonding pairs so the shape is a pyramid.
- iv PF_5 would have five bonding pairs as each F must form one covalent bond with the P. So the shape must be different.
- b (P is a larger atom and can form a maximum of five covalent bonds. Why this occurs and the resultant shape of such molecules are not part of this course.)

Question set 6.2

- 1 Oxygen, nitrogen, fluorine, chlorine
- 2 a There differences in the electrical charge over the molecule; i.e. one 'end' is positive and one end is negative; this is called a dipole.
- b The covalent bond has polarity; i.e. one end will be positive and one end negative.
- c Has different electron charge over the molecule: there is a polar covalent bond present in the molecule and there is asymmetry so the molecule is polar.
- 3 a O-H: O more negative δ^- , H more positive δ^+
 Cl-H: Cl δ^- , H δ^+
 C-H: very small difference, C δ^- , H δ^+
 P-H: no difference
 N-O: O δ^- , N δ^+
 N-Cl: no difference
 N-H: N δ^- , H δ^+
 S-H: S δ^- , H δ^+
 C-S: no difference
 F-O: F δ^- , O δ^+
 Cl-O: O δ^- , Cl δ^+
 C-O: O δ^- , C δ^+
 P-O: O δ^- , P δ^+
 C-Cl: Cl δ^- , C δ^+
- b Highly polar: O-H, N-H, Cl-H, C-O, P-O, C-Cl
 Some polarity: N-O, N-Cl, S-H, F-O, Cl-O
 Non-polar: N-Cl, S-H, C-H, P-H, C-S



- iii No polar bonds

v Non-polar



- iii Polar bonds present

v Polar



- iii No polar bonds

v Non-polar



- iii No polar bonds

v Non-polar

5 a Ammonia: Polar. Pyramid with nitrogen in one corner and three hydrogens

b Iodine chloride: Polar. Diatomic molecule. Chlorine has a much higher electronegativity than iodine, so Cl is δ^- , I δ^+ .

c Dichloromethane (CH_2Cl_2): Polar. Tetrahedral structure with polar covalent bonds between chlorine and carbon and asymmetrical.

d Nitrogen trichloride: Polar. Pyramid shape – slight differences between nitrogen and chlorine in electronegativity and asymmetrical.

e Sulfur difluoride: Polar. V-shape like water, and S-F covalent bond very polar.

6 Hydrogen peroxide is tetrahedral at the two oxygens. It is a bent molecule. When written on a piece of paper the electron dot diagram or valence structure looks like it should be non-polar, like carbon dioxide. But it is a bent molecule with the hydrogens at effectively right angles above the page. Due to the asymmetry, the side of the molecule with the hydrogens is slightly positive (the side above the page) and the other side near the bond of the oxygen (O–O) is negative.

Question set 6.3

- 1
 - a The three strong forces are metallic, ionic and covalent.
 - b The three weak forces are dipole–dipole, H-bonding and dispersion forces.
- 2 Dipole–dipole interactions are caused by the attraction between slightly positive and slightly negative parts of different molecules or atoms. Dispersion forces are due to attraction of the temporary instantaneous dipoles produced due to the movement of electrons in different molecules or atoms.
- 3 NH, OH and FH are the common groups that exhibit hydrogen bonding because hydrogen bonding occurs between the H atom of a NH, OH or FH group to the lone pairs of the N, O or F atoms.
- 4 Water expands when it freezes so if the bottle is full there will be no room for expansion and the bottle will break.
- 5 Dispersion forces are the weakest because they rely on temporary instantaneous dipoles. Dipole–dipole are the next strongest. These forces rely on the attraction of permanent but only slightly charged dipoles. Hydrogen bonds are the strongest due to the permanent dipoles, which have a strong polarity difference. This is due to the difference in electronegativity between hydrogen and oxygen/nitrogen/fluorine atoms.
- 6 The two factors influencing a molecule's polar character are the presence of a covalent polar bond and asymmetry in the shape of the molecule, which results in one end being slightly negative and one end slightly positive.
- 7 Water would boil at less than -100°C as extrapolation on the graph in Figure C6.14 shows.
- 8 H_2S is a heavier molecule with more electrons. As there are more electrons present the chance for dispersion forces increases. This would mean a higher boiling point than a substance with fewer electrons. However, water has hydrogen bonding as well as dispersion forces so it requires more energy to break the intermolecular forces in water. This results in water having a higher boiling point compared to H_2S .
- 9 Liquid ammonia molecules have a slightly higher specific heat capacity than water as ammonia is more polar than water. Specific heat capacity is amount of energy required to heat 1g of substance 1°C . Ammonia molar mass is 17 compared to water's 18 molar mass. So 1 g of ammonia would contain slightly more molecules than water.
- 10 The warmer water is generally found at the surface. Oxygen has to dissolve in from the surface of the water so it is not surprising that the highest levels are found at the surface. The oxygen will only get to the colder lower parts due to the currents and mixing of the cold and warmer water.
- 11 Students' answers will vary. One example is bandages that could be removed without pain.

Question set 6.4

- 1 See Figure C6.22.
- 2
 - a $\text{LiBr} \cdot 2\text{H}_2\text{O}$
 - b $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$
 - c $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$
- 3 Methane has dispersion forces and is non polar. Hydrogen bromide is polar. There is a polar covalent bond and asymmetry present in the molecule. Being polar attracts it to water molecules.

- 4 Sucrose (sugar) dissolves in water. Sucrose has a lot of OH groups which can form hydrogen bonds to the water molecules. The water is the solvent and the sucrose molecules the solute.
- 5 a The salt KBr consists of two ions, the cation K^+ and the anion Br^- . When this salt is placed in water the water molecules will hydrate each ion, thereby the two ions dissociate from each other as the salt dissolves.
- b $KBr(s) \xrightarrow{H_2O} K^+(aq) + Br^-(aq)$
- c An electrolyte is a solution that can conduct electricity. As there are free charged particles (the ions) this solution would be called an electrolyte.
- 6 When the solute ethanol is placed in the solvent water the dispersion forces and hydrogen bonding between the ethanol molecules and between the water molecules are broken. New intermolecular bonding occurs between the ethanol and the water molecules.
- $CH_3CH_2OH(l) \xrightarrow{H_2O} CH_3CH_2OH(aq)$
- 7 a NaCl
- b Polar molecules
- c Some slightly soluble
- d Benzene, petrol, butane
- e Cellulose, plastic polymers
- f A few soluble
- g Covalent lattices
- h Insoluble
- i Insoluble
- j Fe, Al, Mg
- 8 a HBr: Soluble. HBr ionises in water so will dissolve. Will form ion-dipole bonds
- b $MgCl_2$: Soluble. Will form ion-dipole interactions
- c CS_2 : Not soluble. Non-polar
- d $ZnCl_2$: Soluble. Will form ion-dipole interactions with water
- e H_2SO_4 : Soluble. Will ionise and form ion-dipole bonds with water
- f $SiCl_4$: Not soluble. Although the bonds are polar, the symmetry of the molecule means that the molecule is not soluble.
- g Glucose (a sugar, $C_6H_{12}O_6$): Soluble. OH groups are present, which will form hydrogen bonds with water so that the sugar dissolves in water.
- h C_8H_{18} : Not soluble. Only dispersion forces could be formed.
- i NH_2CONH_2 : Soluble. NH groups are present, which will form hydrogen bonds with water so that urea is soluble in water.

- 9 Error 1: The $\frac{1.65}{120} : \frac{3.228}{138}$ ratio should be the ratio of MgSO_4 to H_2O .

The molar mass of anhydrous magnesium sulfate is 120. The 3.228 is meant to be the water of crystallization. This mass should have been divided by 18 the molar mass of water. This would then give the simplest ratio between the salt and the water of crystallization. Instead he assumed that there was only one water molecule as shown by the equation: $120 + 18 = 138$. He then incorrectly divided by his assumed molar mass of the hydrated salt.

Error 2: $\frac{1.65}{120} = 0.01375$, not 0.13749. Was this a miscalculation, or did he not write the full weight measurement off the balance?

Error 3: The simplest whole number ratio would be $\frac{0.13749}{0.02339} : \frac{0.02339}{0.02339}$, or 5.88 : 1.

Error 4: The correct figure is $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. This would mean that if 1.65 was the final weight the n of magnesium sulfate 0.01375. The number of moles of water would then be 7 times this or 0.09625. The mass of water would then be 1.7325. If you add 1.65 to 1.7325 you would get a number close to 3.38. This means the first error was that the 3.228 was probably the initial weight and that he did not subtract the final weight.

- 10 Enzymes denature at high temperatures as there is enough energy to break the dispersion, ion dipole or hydrogen bonding holding the original shape. For enzymes to still function at temperature above 60°C the bonding between the amino acids need to be stronger. This could be achieved by having significantly more bonds or the presence of more covalent bonds (such as the S–S bonds between cysteine amino acids).

Worked example 6.1

- a i 30 g
ii 40 g
iii 80 g
- b NaNO_3 , because it has the highest value
- c At 30°C there would be 36 g of KCl and 36 g of NaCl. The solutions would be saturated as the curve represents where the solutions are saturated.

Experiment 6.2: Deriving solubility curve of ammonium chloride

Discussion

- 1 The solubility of ammonium chloride increases when the temperature increases.
- 2 'Saturated' means the solvent has the maximum amount of solute that can dissolve at that temperature.
- 3 The answer will depend on students' curves. It would be equivalent to 20 g in 100 g of water and should be less than zero.
- 4 At 90°C a saturated solution of ammonium chloride would be around 70 g. At 10°C a saturated solution of ammonium chloride would be around 35 g. So if a saturated solution at 90°C cooled to 10°C , $70 - 35$, or 35 g, would crystallise out.
- 5 Answers will depend on students' experiments.
- 6 Answers will depend on students' experiments.

Taking it further

Answers will depend on students' experiments.

Worked example 6.2

$$\text{a } \frac{6\text{g}}{100\text{ mL}} \times 200\text{mL} = 12\text{ g of sugar}$$

$$\text{b } \frac{36}{300} \times 100 = 12\% \text{ v/v}$$

$$\text{c } 500\text{ ppm} = 500\text{ mg L}^{-1}$$

Convert L to mL:

$$500\text{ mg L}^{-1} = \frac{500\text{mg}}{1000\text{mL}} = 0.5\text{ mg mL}^{-1}$$

Calculate for 100 mL: $0.5 \times 100 = 50\text{ mg}$. 50 mg would be present in 100 mL.

Question set 6.5

1 Saturated; unsaturated

$$2 \quad 30\text{ g L}^{-1} = \frac{30\text{ g}}{1000\text{mL}} = 0.03\text{ g mL}^{-1}$$

a For 50 mL there would be $0.03 \times 50 = 1.5\text{ g}$.

b For 300 mL there would be $0.03 \times 300 = 9\text{ g}$.

3 There is 3.2 g in 40 mL.

Percentage concentration is $\frac{3.2}{40} \times 100 = 8\% \text{ w/v}$.

4 a If 500 mL has 0.3 mg, then 1000 mL would have 0.6 mg. So the concentration is 0.6 mg L^{-1} . This needs to be converted to ppm. $1\text{ ppm} = 1\text{ mg L}^{-1}$. The concentration is 0.6 ppm.

To convert to % w/v: first convert to mL by dividing by 1000 and then multiplying by 100 : $\frac{0.6}{1000} \times 100$
 $= 0.06\% \text{ w/v}$, or, 0.6 mg L^{-1} is 0.0006 g per 1000 mL: $\frac{0.0006}{1000} \times 100 = 0.06\% \text{ w/v}$

b The level is below the maximum recommended levels, because 0.6 is less than 1.

5 At 20°C , 30 g of potassium nitrate would saturate 100 g of water. So 15 g of potassium nitrate should saturate 50 g of water.

6 a No. 2 cm^3 of water is 2 g of water. This would mean $2 \times 1.2\text{ g}$ of sodium thiosulfate could dissolve.

b Saturated

c The solution could be saturated so no more could dissolve.

d The solution would be supersaturated, which means that it is unstable. The addition of a crystal would cause the excess sodium thiosulfate to crystallise out.

$$7 \quad \frac{0.5\text{g}}{20\text{g}} \times 100 = 2.5\% \text{ w/w}$$

$$\frac{0.5\text{g}}{20\text{g}} = 0.025\text{ g/g} = \frac{25\text{mg}}{0.001\text{kg}} = 25\,000\text{ ppm}$$

8 Answers will depend on the sport drinks selected.

9 a $\frac{0.2 \text{ mg}}{200 \text{ mL}} = \frac{1.0 \text{ mg}}{1000 \text{ mL}} = 1.0 \text{ mg L}^{-1} = 1.0 \text{ ppm}$

b No

c Assume cup has 250 mL. Each 200 mL has 0.2 mg. So 250 mL would have $\frac{250}{200} \times 0.2 \text{ mg} = 0.25 \text{ mg}$.

If the daily limit is 10 mg then $\frac{10}{0.25} = 40$ cups.

10 Any valid argument that may include the following:

The Australian manufactures limit that gluten free products must have 'no detectable amount' of gluten means that, as technology improves, lower and lower amounts of gluten can be detected. This would mean that the manufacturers may face increasing cost of testing.

It may be that there is always a small amount of gluten present. For both these reasons it may mean that for the Australian manufacturer it would no longer be economical to produce gluten free products. People with coeliac disease may have a more restricted diet.

The British and Europe manufacturers' maximum limit of 20 mg of gluten per kilogram would mean that they would not be facing such increased cost of testing. If this limit is below that which causes problems for the people with coeliac disease then it would not be a problem. Consideration should also be given that the sensitivity to gluten products may not stay at the same rate, so it may be that a lower level is desired.

Experiment 6.3: Thin layer chromatography

Discussion

4 A TLC chromatograph could be run with standards of known materials. If the same R_f was achieved under the same conditions then it would help identify the unknown.

Question set 6.6

1 While both soap and detergent are surfactants, soap has a polar head and a non-polar long hydrocarbon chain typically made from animal or plant fatty acids. Detergents may have a range of different polar end groups. Anionic detergents work in hard water as the sulfate groups do not precipitate the calcium or magnesium ions present, unlike soap, which does.

2 a To identify the samples present, a standard of the suspected drug must be run under the same conditions. Running a commercial drug on the same plate will ensure that the conditions are the same. If the R_f s are the same, the investigators will be confident that the sample contains that drug.

b The sample may be a mixture, containing some of drug Y. The R_f of drug Y and one of the sample components is $\frac{14.6}{20} = 0.73$.

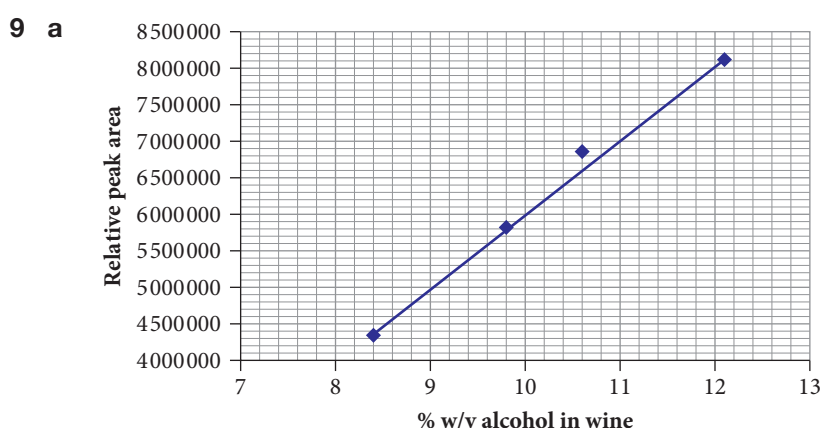
3 The GC consists of a carrier gas bottle, a flow controller, an injection port, an oven with a column inside, a detector and a recorder.

4 Standard known compounds are run under the same conditions. If the R_t is the same then the component can be identified for both GC and HPLC.

5 A sample is injected into the heated injection point. Here it is moved into the column, which is in an oven. The carrier gas is the mobile phase. The solid phase is in the column. The components move through the column. The components of the sample will have differing rates of adsorbing onto the

stationary phase or desorbing into the mobile phase. Smaller and less attracted to the stationary phase components will pass through the column quicker; they will elute in a shorter time. These organic substances will have a shorter R_f . The organic substances which are larger or are more strongly attracted to the stationary phase will elute in a longer time. They will have a larger R_t . When the samples elute they are detected and a chromatogram is produced that records the peak area response of the detector against the time to elute.

- 6 Smaller samples elute quicker so compound 1 is likely to be the smallest.
- 7 The sample is not pure. It appears that there is one main component that has an R_f of 4. It is likely to have the highest concentration and be the main ingredient. There is also a range of smaller organics present.
- 8 a The sample that spent most time in the stationary phase is the one with the highest R_f . This would be D.
b C percentage is determined by estimating the ratio of the height of C against the total of the others.
 $\% C = 7 / (6 + 10 + 7 + 9) \times 100 = 22\%$
c Honey decomposes when heated over 40°C , so the HPLC is preferred to GC in this analysis.
d To confirm the identity of peak C, a sample of pure fructose would be run under the same conditions. If the same R_t was obtained it would verify that the peak was fructose.



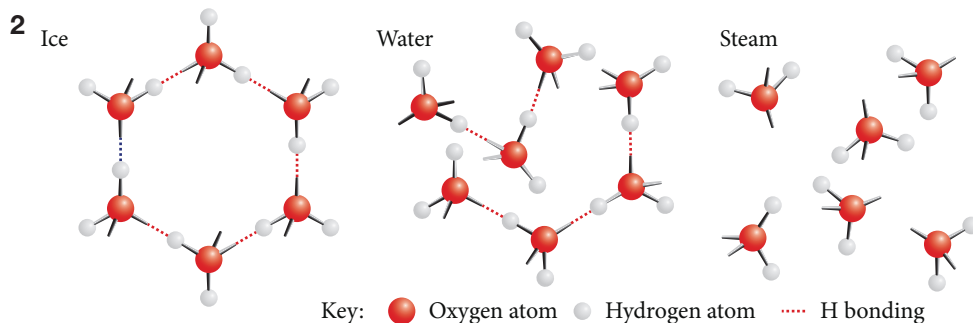
9.32% w/v

- b $9.32\% \div 0.785 \text{ g mL}^{-1} = 11.9\% \text{ v/v}$
 - c A calibration curve means that you use the line of best fit. Each point will have a small error so by using the line of best fit the effect of these errors is reduced. If you just used the ratio of the peak area of the unknown to one standard then the error would be larger.
 - d 7% v/v wine's peak area would be outside of the calibration curve. Extrapolation outside the range of the standards may not be accurate.
- 10 a I, IV, III, II, V, VI, VII, VIII, IX, X
- b Errors: Any three of: i) A pencil should be used not a gel pen. ii) The spot should not be too large. iii) The origin should be above the solvent. iv) The component that adsorbs most strongly will travel the shortest distance.
 - c R_f s are calculated by measuring the distance the component has moved and dividing it by the distance the solvent has moved (the distance from the origin to the solvent front).

- 11 Students' answers may vary, such as: With the development of HPLC and GC the authorities can better detect drugs and other banned substances that may be used by athletes. This means more sport cheats could be caught. This may push the boundary of what other substances would be used. The legal system needs to keep up with the developments.

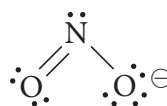
Chapter review questions

- 1 Covalent bonding within the molecule and dispersion forces between the molecules.

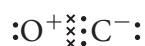


3 B

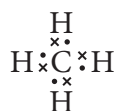
- 4 a 3 bonding pairs and 6 non-bonding pairs



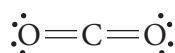
- b 3 bonding pairs and 2 non-bonding pairs



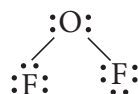
- c 4 bonding pairs and no non-bonding pairs



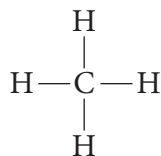
- 5 a Non-polar



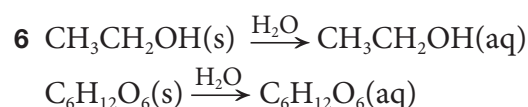
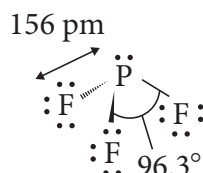
- b Polar



c Non-polar



d Polar



7 When potassium iodide dissolves in water the ionic bonds between the potassium and iodide are broken. This is due to the water molecules moving in between the lattice. The water molecules then form ion-dipole bonds with the potassium ions and with the iodide ions. These ions are hydrated by the water molecules. The oxygen end of the water molecule is orientated towards the potassium ion. The hydrogen end of the water molecules orientate towards the iodide ion. The ions are no longer in the lattice, they are dispersed into the water, dissolved.

8 The formamide liquid would be drawn up the glass capillary tube.

9 D; longer retention time is due to moving slower through the column. So the component would be larger or more strongly adsorbed onto the stationary phase.

10 The intermolecular forces between the liquid molecules will generally occur in three dimensions. This means that a molecule in the middle of the liquid will be pulled in all 3 dimensions. However at the surface of the liquid the molecule will only 'feel' the intermolecular forces from molecules within the liquid. This pulls the molecules towards the centre of the liquid. It is this that causes the surface to be under tension.

11 a Place a drop of diesel and a drop of oil onto a glass plate. Make sure the surface is clean and dry beforehand. Measure the contact angle.

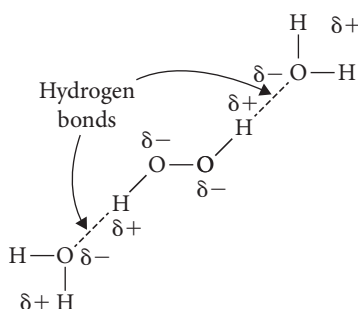
b The material with the greatest surface tension will have the smallest contact angle.

c This is because the material with the greatest surface tension will pull the molecules of the liquid together and have a lower contact to the surface.

12 A needle can float on water as the weight of the needle is supported by the tension in the water. However when a drop of detergent is added, the surface tension is reduced. The needle will no longer float.

13 Total volume = $375 \times \frac{8.5}{100} = 31.9 \text{ mL}$

14 Hydrogen peroxide is polar so it would be expected to be soluble in water.



15 a Potassium nitrate will form a saturated solution when 30 g of potassium nitrate is dissolved in 100 g of water at 20°C.

b At 30°C and 50 g of water, 23 g would be required to make a saturated solution.

c At 10°C and 100 g of water 20 g of potassium nitrate would be required to make a saturated solution.

So 80% of this is required to saturate 80 g of water or $20 \times \frac{80}{100} = 16$ g.

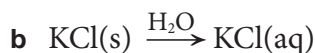
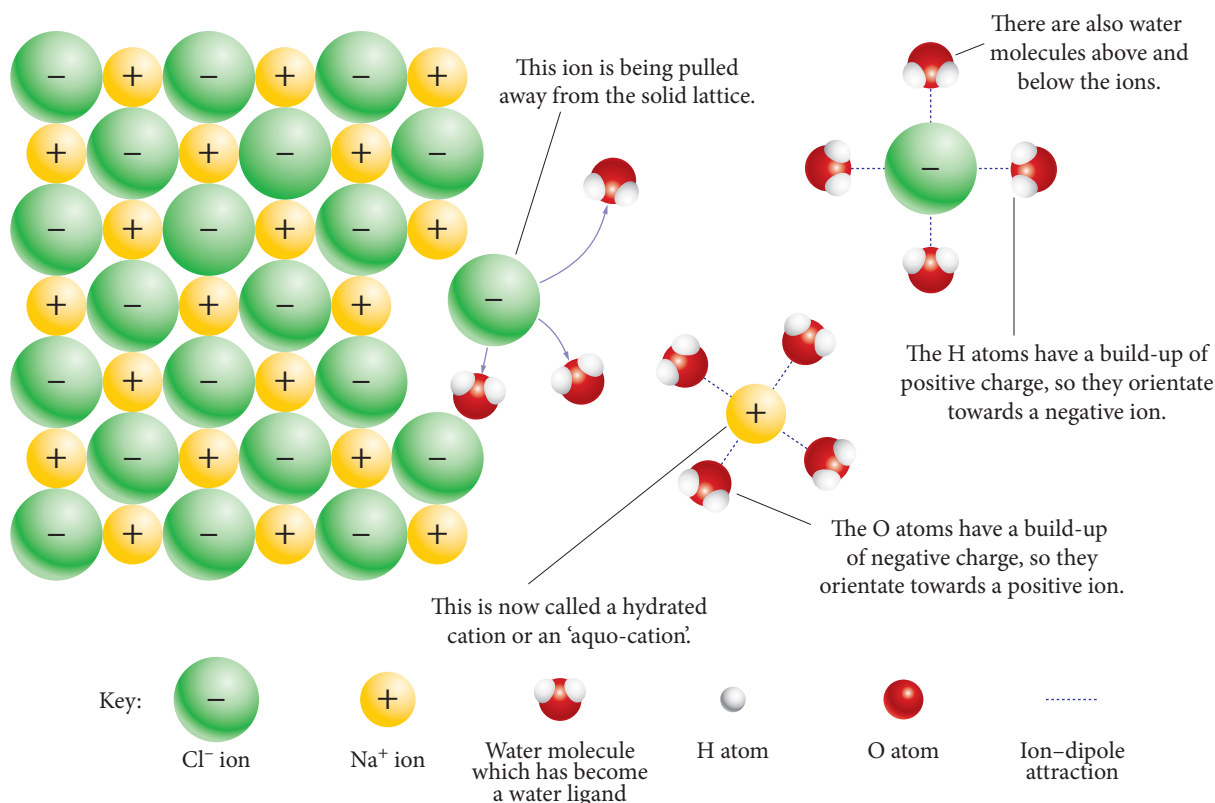
16 Serotonin concentration is $1.5 \text{ ng L}^{-1} = 1.5 \times 10^{-9} \text{ g L}^{-1}$ (1 nanogram is 10^{-9} g). This is the amount in 1 litre so the amount in 1 mL is $1.5 \times 10^{-9} \times 10^{-3} = 1.5 \times 10^{-12}$ g.

17 A. A water in oil emulsion will not readily mix with water, nor will conduct electricity and cannot be coloured by the addition of a water soluble dye. These emulsions feel greasy. They will feel and behave more like an oil than water.

18 Table 6.2 shows that as the size of the noble gas atoms increase, the boiling point increases. i.e He has a boiling point of -269°C and Radon has a boiling point of -62°C . As the size of the atom increases there are more electrons present so that there are more chances for the temporary dispersion forces to occur. With more dispersion forces or intermolecular forces the energy required to break the forces also increases.

The same trend is seen in the Halogens with Fluorine boiling point of -188°C and Astatine boiling point of 337°C . Astatine has more electrons so more chance for dispersion forces than Fluorine.

19 a



- c Reading off Figure C6.25: at 80°C , 100 mL of water would dissolve about 52 g of KCl. So 50 mL of water would dissolve about 26 g of KCl.
- d If this was then cooled to 10°C the solution would be supersaturated, as at 10°C a saturated solution would have 15 g of KCl.
- e 100 g of water will dissolve 40 g of KCl to make a saturated solution when it is heated to around 42°C . If this was cooled to 10°C then $40 - 30$ g of KCl or 10 g would drop out of solution.

20 400 mg magnesium hydroxide is in 10.0 mL

As % w/v: $0.400 \text{ g}/10.0 \text{ mL} \times 100 = 4\% \text{ w/v}$

As ppm: $400 \text{ mg in } 10 \text{ mL} = 400 \text{ mg}/(10/1000 \text{ L}) = 40\,000 \text{ mg L}^{-1} = 40\,000 \text{ ppm}$

- 21 Each solute has a different molecular structure. So each component has a different size and shape. The atoms present and the shape of the molecule will also affect the type and number of intermolecular bonds that may form. These intermolecular bonds will occur between the molecule and the stationary phase and to the mobile phase. These factors all affect the relative time each component spends adsorbed onto the stationary phase and time it desorbs back into the mobile phase. Therefore different solutes travel through a gas chromatograph at different rates.
- 22 Blood's main component is water. As steroids do not circulate in the blood by themselves it would mean that steroid hormones are non polar. However as dopamine can travel through the blood system it would indicate that dopamine is polar. Dopamine is more likely injected directly into the veins as it can dissolve in the blood. If a substance that was non polar was injected then the vein would get a blockage.
- 23 Methane is a tetrahedral molecule. Ammonia has a pyramid molecule. In methane the electronegativity of the C and H are similar so that the bonds are non polar. In ammonia N is much more electronegative than H and the covalent bonds present are polar.

Methane is a symmetrical molecule with no polar bonds. Only dispersion forces will exist between the methane molecules.

Ammonia is an asymmetric molecule with polar bonds. The apex of the pyramid will be slightly negative due to N's higher electronegativity. The base of the pyramid will be slightly more positive. This means that H bonding can occur between ammonia molecules as well as dispersion forces.

As ammonia can form H bonding with other ammonia molecules it can also form H bonding with water molecules. For this reason ammonia is soluble in water.

As methane can only form dispersion forces it is non soluble in water.

24

General formula	Lone pairs	Bonding pairs – lone pairs	Shape
MX	0	1	linear
MX ₂	0	2	linear
MX ₃	0	3	trigonal planar
MX ₄	0	4	tetrahedral
MX ₆	0	6	octahedral
MX ₃ E	1	3	trigonal pyramidal
MX ₂ E	1	2	V shape
MX ₂ E ₂	2	2	V shape

25 The students may have many responses.

Originally the structure of the molecules was thought to be mostly due to the valence bonds or how the atoms/ligands pack around the central atom. The final shape was thought to rely upon the minimisation of the repulsion between the outer ligands. This explains the tetrahedral structure of methane. Then further research into molecular structures would be undertaken. Ammonia would be determined to have a pyramidal geometry. This could not be explained by the original theory. The influence of lone pairs in determining the shape of the molecule could be explained by the VSEPR model. Research would continue to determine what was the most important factor affecting the shape of the molecules. Data from many molecules would be collected. It would be observed that the VSEPR model could explain a wider range of molecules and shapes. The model would be refined over time, the strength of repulsion between lone pairs, and between lone pairs and a bond pair would be determined. The influence of single, double and triple bonds would be determined. This model would be used to predict molecular structures. The VSEPR model was also used to determine reaction rates. It would be found to accurately predict the shape and angles for many molecules.

As technology grows the ability to refine the theory and test the limit of the theory would also extend. It would get to the point where the theory would have a significant number of exceptions. For the VSEPR theory it is the neglect of the ligand- ligand repulsion. From this understanding a better more accurate model is developed and used by the researchers to develop an even better understanding of more complex molecules structures and reactions. The new model incorporates effects due to the close packing of ligands. The researchers then go on to test the limits of this model.

From research, experimental data is collected. A model or theory is developed that can explain the current results. Further research is undertaken to test the theory or model. The model/theory limits on what it can be applied to is found. The model is changed, the model is further tested and refined and developed over time. The model helps other researchers in their fields. The VSEPR theory is a good 'back of the envelope' prediction of the shape of molecules for researchers and for teachers!